SOVEREIGN INVESTOR DASHBOARD · SLOVAKIA & CZECHIA

Full Audit — Features, Bugs, Gaps, Manual Checks & APIs

Every feature mapped · bugs ranked by severity, each with its fix · what only you can check · what is missing · what could be added · where to get APIs

3,850	0	29	3
TESTS PASSED	TESTS FAILED	FINDINGS	HIGH SEVERITY

Review date: 12 November 2026 · Code reviewed: `main` at commit `ad3f4d9` (after PR #76) · Read-only review: nothing in the app's code was changed. Your real database and listings were never touched; the app was run on a scratch copy with test data.

The short version. The code is in good shape: all tests pass, no crash and no script injection turned up in a real browser, no secrets or vulnerable packages. All but three of the last audit's findings are closed. What is left are three things that matter: **(1)** the Slovak title-deed screen still calls a deed clean when it holds an entry it has no rule for; **(2)** the Telegram deal alerts will almost never fire with the default settings; **(3)** the tax tables run out on 1 January 2027.

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1. Every finding at a glance

Severity. HIGH can lead to a wrong buy/skip decision, or a feature you rely on doesn't deliver in normal use. MEDIUM gives weaker or misleading results in some cases, or is a risk that grows. LOW is a rare edge case, cosmetic or housekeeping. INFO is a note. **Effort:** S = under half a day · M = a few days. ✔ = reproduced with a test during this review. **Who:** "code" can be fixed for you; "you" needs you or a professional.

#	Severity	Finding	Where	Who · effort
1	HIGH	✔ The Slovak title-deed (LV) screen calls a deed **clean** when part C holds an entry it has no rule for. Thirteen real wordings passed, among them a lifetime right to live in the flat, a pending change of owner (plomba) and a restriction on disposal.	`modules/lv_screen.py`, `debt_bot.py`	code · S–M
2	HIGH	✔ **Telegram deal alerts will almost never fire.** They require a flat that isn't cash-flow negative, and with the default financing that needs a price 46–55 % below the median — where the 50 % sanity floor already treats the price as a misread.	`modules/alerts.py`, `config.py`	code · S
3	HIGH	The tax tables expire on **1 January 2027**, seven weeks away. No 2027 table exists for either country. Open since two reviews ago.	`config.py`, `markets.py`, property-tax engines	you + code · M
4	MEDIUM	The s.r.o. tax model misses three rules: dividend tax is 7 % on profits from 2025 (the model uses 10 %), the minimum corporate tax of €340 a year is not charged, and depreciation of the flat is not deducted.	`config.py`, `engine/financial.py`	you (účtovník) + code · S
5	MEDIUM	✔ The dashboard shows the **s.r.o.'s** figures by default, while the engine recommends Personal for almost every flat. Surplus, the ⛔ flag, the "cash-flow positive" filter and the +2 pp stress all follow the s.r.o.; the alerts use the better structure, so the two can disagree.	`dashboard/sidebar.py`, `view.py`, `cards.py`	code · S
6	MEDIUM	✔ The duplicate matcher merges **different flats**. Three Petržalka flats on floors 2, 7 and unknown became "one flat on 3 portals"; two were hidden.	`engine/duplicates.py`	code · S–M
7	MEDIUM	On a Mac without Docker nothing runs by itself: the launcher starts only the dashboard. No 06:00 run, no backup, no alerts unless you click ▶ RUN EVERYTHING.	`Start Sovereign.command`	code · S (you: workaround)
8	MEDIUM	An LV **PASS never expires** and its date isn't on the card. Only PENDING and UNVERIFIED deeds are re-checked, and a cached deed can be 30 days old.	`database/title_deeds.py`, `cards.py`	code · S
9	MEDIUM	✔ Some banks aren't recognised, so their ordinary mortgage rejects the flat: Privatbanka, the abbreviation SLSP, and older names still on deeds (Istrobanka, Volksbank, Sberbank, Citibank).	`modules/lv_screen.py`, `config.LV_BANK_NAMES`	code · S

10	MEDIUM	✔ Villages and city parts outside the lookup get **no price benchmark** and the national default rent, so they can never be GREEN (Dunajská Lužná, Košická Nová Ves, Trenčianske Teplice, Veľký Meder).	`engine/regional_prices.py`, `financial.py`	code · M
11	MEDIUM	Backups still sit on the same disk as the database. A restore overwrites the live database without first saving it.	`modules/backup.py`	you · S, code · S
12	MEDIUM	Seven Slovak towns still use the €1.35/m² property-tax fallback (their sites block automated reads).	TAX RATES tab	you · S
13	LOW	✔ Any title containing “novostavba” is flagged a developer project, hidden by default and never alerted — also a single resale flat in a new building.	`database/listings.py`	code · S
14	LOW	✔ The CSV export writes cells starting with `=`, `+` or `@` as they are; Excel runs them as formulas. An ad title is text a stranger wrote.	`dashboard/tabs/triage.py`	code · S
15	LOW	✔ The map opens on the average of all pins, so one wrong pin moves it off the listings (the test map opened over Ukraine). Coordinates are validated in the scrapers only.	`dashboard/tabs/map_view.py`	code · S
16	LOW	A Telegram message lists every price cut and problem with no cap. Past 4,096 characters Telegram refuses it, and it is then retried and refused on every run.	`modules/alerts.py`	code · S
17	LOW	A corrected size or room count on a listing doesn't re-score it; only a new price or district does.	`database/listings.py`	code · S
18	LOW	The Google integration calls the legacy Places endpoint. Google no longer lets new projects enable it, so a fresh key may not work (the app then falls back to OpenStreetMap).	`modules/location_iq.py`	code · S–M
19	LOW	✔ `analyse()` crashes when the district is `None`. The pipeline always passes text, so it can't happen there.	`engine/financial.py:621`	code · S
20	LOW	Login: a page refresh asks for the password again, five wrong tries lock everyone out for 15 minutes (also you), and on a network the password travels without HTTPS.	`dashboard/login.py`	you · S
21	LOW	The page header (banner, tiles, health strip, notes) takes about 800 px before the tabs start; on a laptop every tab begins below the fold.	`dashboard/overview.py`	code · S
22	LOW	✔ “Košice - Staré Mesto” is priced at Košice II's rent (€12.00/m²), not Košice I's (€13.50): about 11 % low.	`engine/financial.py`	code · S
23	LOW	The Slovak mortgage rate (3.8 %) and term (25 years) can't be set in `.env`; the Czech ones can. Scores, the triage table and alerts use them.	`config.py`	code · S

24	LOW	Slovak cooperative flats (družstevný byt) aren't recognised; the Czech ones are. Such a flat is cheaper, has no deed of its own and is hard to mortgage, so it can look GREEN.	scrapers, `debt_bot.py`	code · S
25	LOW	Ad text is sent to Claude for parsing. A crafted ad could nudge the parsed features (parking, furnished, condition), which move rent by a few per cent. Nothing else is reachable.	`modules/llm_enrichment.py`	—
26	INFO	Two credentials were pasted into this chat: the Telegram bot token and an Anthropic credential. Neither is in the repository or its history. Revoke both (manual task A).	—	you · 10 min
27	INFO	✔ At the default financing a flat at the median price loses €290–460 a month; gross yields are 3.6–4.2 %. This is the market, not a bug, and it is the reason behind finding 2.	engine	—
28	INFO	109 broad `except Exception` handlers (104 last time), all logging what failed.	codebase	—
29	INFO	362 deprecation warnings per test run, all from inside Streamlit (pandas 4). Pinned below pandas 4 already.	dependency	—

If you do only three things: (1) let me fix findings 1 and 2 — until then read every Slovak LV yourself and don't wait for a Telegram message; (2) book an účtovník for the s.r.o. questions (finding 4) and the 2027 figures (finding 3); (3) revoke the two pasted credentials and put fresh ones in `.env` (manual task A).

2. Everything the dashboard can do — the feature map

Sovereign finds flats for sale in Slovakia and Czechia priced below the local market, works out their cash flow under each country's tax and lending rules, screens their title deeds, tells you on Telegram, and tracks the ones worth pursuing through to a purchase-contract draft. It is **two programs on one SQLite database** (`data/sovereign.db`): the **dashboard** (`app.py`) and the **scheduler** (`scheduler.py`, daily at 06:00 Bratislava time). **New since the last audit** is marked ★.

2.1 How it runs

Way	What it starts	Notes
Windows + Docker (`START.bat`)	Dashboard and scheduler	Port bound to this computer only. ★ Runs as an unprivileged user.
Mac without Docker (`Start Sovereign.command`)	**Dashboard only**	No scheduler (finding 7). Python 3.11–3.14.
Manual (`restart.sh`, `python scheduler.py`)	Either	Bound to 127.0.0.1
Password gate (`DASHBOARD_PASSWORD`)		★ Locks for 15 minutes after 5 wrong passwords

2.2 Markets

	🇸🇰 Slovakia	🇨🇿 Czechia
Sale portals	Nehnutelnosti.sk (headless Chromium), Bazoš.sk, Topreality.sk	Bezrealitky.cz, Reality.iDNES.cz, Bazoš.cz (Sreality left out: robots.txt)
Rental portals	Bazoš, Topreality	Bazoš, Bezrealitky
Currency	EUR	CZK, converted at the ČNB rate fetched daily
Price benchmark	Regional median €/m², room-adjusted, per okres in Bratislava	ČSÚ purchase prices, live Prague and Brno district levels
Rent baseline	`config.RENT_PER_M2`, about 110 towns; kraj median and national default as fallbacks	Deloitte Rent Index; ★ the Ministry of Finance rent map for about 4,800 more municipalities
Title deed (LV)	kataster.skgeodesy.sk (unofficial scraper)	ČÚZK API KN (free key) plus a pasted výpis z LV
Property tax	Per-town rate per m², read from the towns' own sites	Coefficients from each ordinance (6,258 municipalities)
Lending rules	NBS: 80 % LTV, 70 % for a 3rd+ property; +2 pp stress	ČNB: 80 %, 70 % for an investment mortgage
Personal tax	★ 2026 bands 19/25/30/35 %, on top of your other income; ★ optional interest deduction	§ 9: flat-rate or actual expenses, 15 % / 23 %

2.3 The daily pipeline (14 steps in Slovakia, 13 in Czechia)

#	Step	What it does
1	Backup	A copy of the database and a readable JSON of everything you typed; the newest 14 are kept
2–4	Portal scrapes	Up to 10 pages per portal; each portal's count is recorded. A portal that returns nothing fails its step and keeps its listings.

5	Fill missing details	Re-reads up to 100 listings without a price, size or district
6	Rent comps	Rental ads become a live rent per m² per district, blended with the baseline
7	Housekeeping	Retires listings unseen for 21 days, flags developer projects, ★ deletes title-deed texts older than 180 days
8	LV debt filter	PASS, REJECTED or UNVERIFIED per flat. ★ Knows transfer bans and the owners' statutory lien. Claude reads the deed too when a key is set; ★ the dashboard says when it isn't.
9–10	Address and description parsing	District, parking, balcony, furnishing, lift, condition (rules, plus Claude)
11	Cash-flow scoring	The financial engine on every unscored listing
12–13	Location IQ, risk backfill	Geocoding; transit, amenities, flood, noise and construction from open data
14	Merge portal copies	One row per flat across portals
★	Telegram alerts	After every full run: new deals, price cuts on deals you track, failed runs and quiet portals

Safety: catch-up after downtime, three hourly retries, one lock shared by dashboard and scheduler, status shown in red when a run failed. Each market runs on its own.

2.4 The financial engine

- **Rent:** live comps or baseline, adjusted for rooms, parking, furnishing, balcony; says when it is a fallback.
- **Costs:** mortgage split into interest and principal, HOA or SVJ, per-town property tax, 5 % vacancy, 5 % reserve, optional management, acquisition costs.
- **Two structures side by side:** personal and s.r.o. ★ The s.r.o. is charged for the extra cash it needs before it can be recommended, and ★ the headline returns are the recommended structure's.
- **Outputs:** NOI, cap rate, yields, cash-on-cash, total ROI, monthly surplus per structure, max offer 🟡/🟢, +2 pp rate shock, 10-year IRR, equity multiple.
- **Class:** GREEN at 20 % or more below the regional median, YELLOW at 10 %, WHITE otherwise; a price over 50 % below is treated as a misread. **Grade A–D** from yield, discount, location, energy, condition and risk.
- Scores are redone when the tax year, a price, a district's rent, a town's tax or the model version changes.

2.5 Page header

- Market switch (All, Slovakia, Czechia), six KPI tiles, price cuts in the last 14 days.
- ★ **Portal health strip:** each portal's latest count, a 7-day mini chart and its last run; red when it failed, amber after 36 hours.
- Banners: quiet portal, Czech listings waiting for a rent, stale tax year, old FX rate, failed run, ★ keyword-only LV checks.

2.6 The 12 tabs

Group	Tab	What you can do there

🎯 DEALS	TRIAGE TABLE	Every scored listing, sortable. ★ Hide columns (Grade, Title, Price, Surplus stay pinned). ★ CSV export.
	ACTIVE SNAG LIST	GREEN and YELLOW cards, 25 a page; WHITE as one table. Costs, both structures, risks, price history, portal copies, deal stage, notes, PDF memo, LV re-check. ★ Compact cards for a phone. ★ “Last seen N days ago”.
	MAP	★ Colour by class or cash flow, ★ group nearby pins, ★ show only listings within 1–50 km of an address or coordinates
	SATELLITE VIEWER	The listing photo against satellite and Street View; a vibe score with notes
📊 ANALYSIS	WHAT-IF / TAX	Re-run any listing or your own numbers: price, LTV, rate, term, rent, hold, growth. ★ The hold period moves the structure verdict.
	★ COMPARE	Two to four listings side by side
	RENT COMPS	Live rent against the baseline per district, sample sizes, Czech towns waiting for a rent
	TAX RATES	Each town's rate against its own site, doubtful readings to accept or reject, ordinance PDF upload
🗂 WORKFLOW	DEAL PIPELINE	WATCHING → VIEWING → OFFER → NEGOTIATING → DUE DILIGENCE → NOTARY → CLOSED or PASSED, with notes and a timeline
	LV TO-DO	Deals whose deed isn't verified; bulk entry of LV numbers
	REJECTED	Every LV rejection with its reason and Claude's read; re-verify
	ONE-CLICK CLOSE	A pre-filled kúpna zmluva or kupní smlouva draft to download

2.7 Sidebar

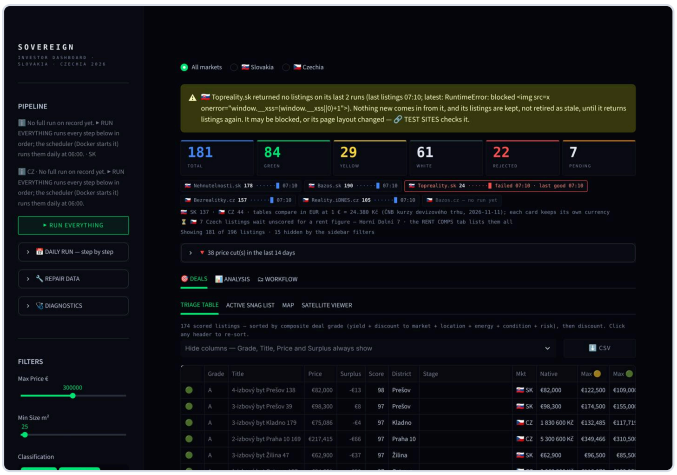
- ▶ **RUN EVERYTHING**; one button per portal and per step; repair tools (rescore all, re-parse, update tax rates, clean stale); diagnostics (test sites, ★ test Telegram).
- **Filters:** max price, min size, class, portals, condition, district, price drops only, hide developer projects, hide duplicates, lift, furnished, parking, cash-flow positive only, demo data, s.r.o. toggle.

2.8 Documents, keys and maintenance

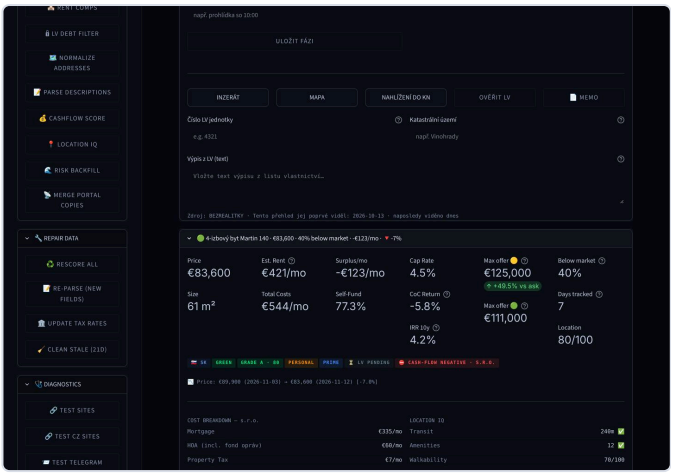
- PDF investment memo, contract drafts (SK, CZ), backups and restore.
- **Optional keys:** Anthropic Claude, Google, ČÚZK API KN, ScraperAPI, ★ Telegram bot.
- **Quality gates:** 3,858 tests in 98 files, ruff, CI on Linux (Python 3.11, 3.12, 3.14) and Windows, a browser job, Dependabot.
- 59 settings in `.env.example`.

2.9 Screenshots from this review

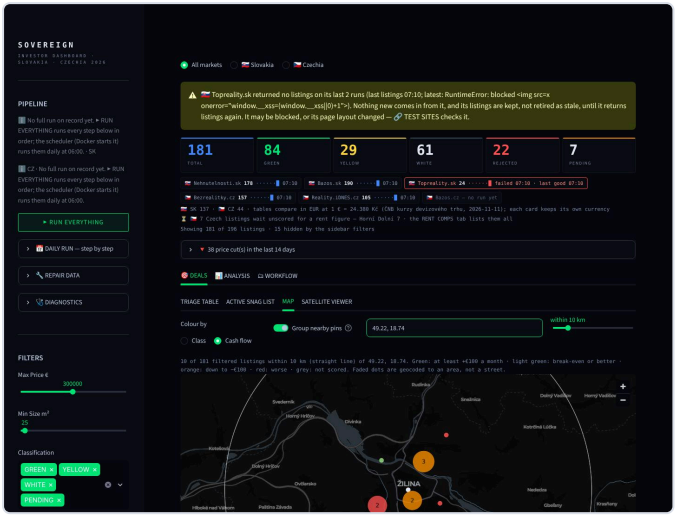
Taken on a scratch copy with 221 test listings. Some carry deliberately hostile text such as `<img src=x onerror=…>`; it shows as plain text, which is the result wanted.



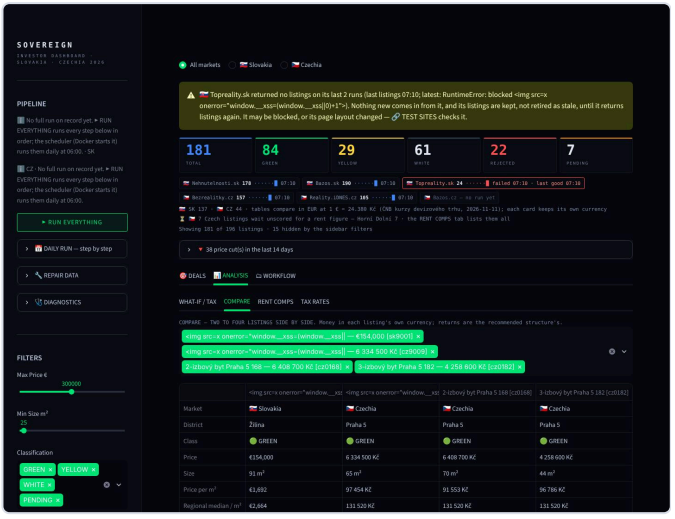
DEALS → TRIAGE TABLE with the health strip, the column chooser and CSV



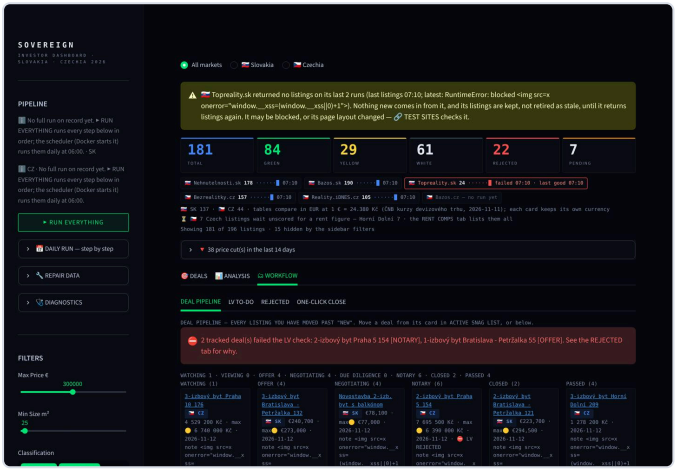
An open card. Note “PERSONAL” recommended, yet “COST BREAKDOWN — s.r.o.” and “CASH-FLOW NEGATIVE · S.R.O.” (finding 5)



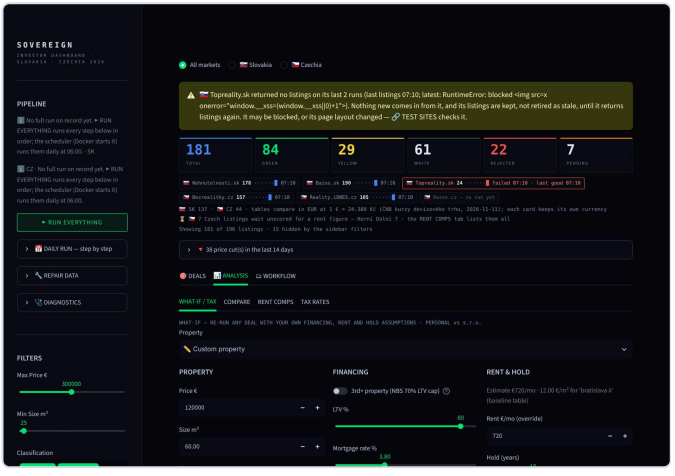
DEALS → MAP: cash-flow colours, grouped pins, within 10 km of a point



ANALYSIS → COMPARE with four listings, two of them hostile test rows



WORKFLOW → DEAL PIPELINE



ANALYSIS → WHAT-IF / TAX. The tabs start about 800 px down (finding 21).

3. Test and check results

Check	How	Result
Full test suite	`pytest` · Python 3.13.18	**PASS** 3,850 passed, 8 skipped (the browser module), 0 failed · 141 s
Browser scraper tests	CI's “browser” job on PR #76	**PASS** green on GitHub, 12 Nov
Lint (CI's rules)	`ruff check .`	**PASS**
Known-vulnerable packages	`pip-audit -r requirements.txt`	**PASS** none found
Secrets in the repository and its 263 commits	`git grep` and `git log -S` for key patterns, the bot token and the chat id	**PASS** none
Dead code	vulture at 80 % confidence	**PASS** none (only cache-key arguments)
Every tab in a real browser	Chromium, all 3 groups and 12 tabs, cards opened, market switched, map modes, pickers, CSV	**PASS** no Python exception, no JavaScript error
Script injection (XSS)	Hostile HTML in titles, districts, addresses, descriptions, LV details, Claude summaries, deal notes, vibe notes, portal errors, rental titles and the map's search box; checked on every tab	**PASS** nothing executed
Phone width (390 px)	Page width, compact cards	**PASS** no sideways scroll
Engine edge cases	28 probes: zero and negative price, zero size, 0 % rate, 100 % LTV, odd room counts, Czech towns, IRR extremes	**1 crash** on a `None` district (finding 19); the rest give a number or refuse by design
Slovak LV screen, crafted entries	33 part-C entries	**13 pass that should not** (finding 1) · **6 reject that should not** (finding 9)
Duplicate matcher	Three look-alike flats on different floors	**merged** (finding 6)
Break-even discount	A 55 m² two-room flat in six towns at default financing	**46–55 % below the median** (findings 2 and 27)
CSV export	Cells starting `=`, `+`, `@`	**written unchanged** (finding 14)
Speed	First load 3.8 s at 221 listings. 0.4–0.5 s a click at 3,000 listings, measured on this code earlier the same day	**PASS**

4. Bugs and fixes in detail, by severity

Nothing was fixed in this pass. Every fix marked “code” can be done for you in a follow-up session.

High

1 · The Slovak LV screen calls a deed clean when it holds an entry it has no rule for

HIGH

`modules/lv_screen.py` (`_RULES`), `modules/debt_bot.py` (`_parse_lv`, `check_flat_lv`) · code · S–M · ✔ reproduced

Problem

The screen looks for the words it knows. A part-C entry or a note worded any other way produces no hit, and no hit means “Clean title — no blocking encumbrances”. The Czech screen has a safety net for this (an entry no rule names makes the deed UNVERIFIED) and rules for a plomba, a lease and a výměnek. The Slovak screen has neither.

Evidence

Each of these, alone in part C, returned **PASS**:

- “Právo doživotného bývania a užívania bytu v prospech Mária Nováková” (a lifetime right to live there, written without the words “vecné bremeno”)
- “PLOMBA VYZNAČENÁ NA ZÁKLADE V-1234/2026”, and a pending “návrh na vklad kúpnej zmluvy” (someone is already buying or pledging it)
- “Duplicitné vlastníctvo”, “Hodnovernosť údajov katastra … bola spochybnená”
- “Obmedzenie nakladania s nehnuteľnosťou” (by the tax office), “Nútená správa”
- “Začaté vyvlastňovacie konanie”, “Povolená reštrukturalizácia”, “ochrana pred veriteľmi – splátkový kalendár”
- “Právo spätnej kúpy”, “Nájomné právo … na dobu 10 rokov”, “Zádržné právo”

Impact

The dangerous direction: a flat you can't freely buy, or that comes with a sitting occupant, shows LV PASS. With an Anthropic key Claude may catch it; without one nothing does.

Fix

(a) Add rules for the thirteen wordings, each with a test. (b) Port the Czech safety net: split part C into its entries, and when any entry matches no rule, return UNVERIFIED with “part C holds an entry the screen doesn't know — read it”. (c) Show part C's text on the card so you can read it in one click. Until then: read every Slovak LV yourself before an offer, PASS or not.

2 · Telegram deal alerts will almost never fire HIGH

`modules/alerts.py` (`qualifying_deals`) · code · S · ✔ reproduced

Problem A deal is alerted only when it is GREEN or YELLOW **and** not cash-flow negative under its better structure. I built the rule that way from the last audit's wording (“a new GREEN, cash-flow-positive listing”). It doesn't fit the model's numbers.

Evidence How far below the median a 55 m² two-room flat must be priced to break even at the defaults (80 % LTV, 3.8 %, 25 years):

Town	Loss at the median price	Break-even at	… with a 30-year loan	… and 70 % LTV
Bratislava-Petržalka	−€461/mo	52 % below	47 %	39 %
Košice	−€402/mo	52 %	47 %	39 %
Žilina	−€292/mo	46 %	40 %	31 %
Nitra	−€303/mo	51 %	46 %	38 %
Prešov	−€307/mo	51 %	46 %	38 %
Martin	−€300/mo	55 %	50 %	42 %

A price more than 50 % below the median is treated as a misread and made WHITE. So the window in which a flat is both GREEN and cash-flow positive is between zero and four percentage points wide. In the test data, 76 of 84 GREEN flats carried the ⛔ flag.

Impact You set the bot up to hear about deals and will hear about almost none. Price cuts on tracked deals and failed runs still arrive. The “cash-flow positive only” filter is nearly empty for the same reason.

Fix Give the alert its own rule, separate from the dashboard's ⛔ flag: by class alone, or by a surplus you choose (for example `ALERT_MIN_SURPLUS=-150`), or by grade (A and B). Also announce a deal again when it moves from YELLOW to GREEN or is cut after its first alert. Stop-gap you can do today: set `CASHFLOW_MIN_SURPLUS=-200` in `.env`; that also moves the dashboard's flag.

3 · The tax tables expire on 1 January 2027 HIGH

`config.TAX_YEAR_RULES`, `markets.CZ_TAX_YEAR_RULES`, the property-tax engines · you + code · M · open since the 5 Nov review

Problem Only 2026 tables exist. From January, scoring falls back to them and the dashboard warns. Bands, the subsistence minimum, lending limits, the Czech inflation coefficient and towns' ordinances can all change on 1 January.

Fix Collect the 2027 figures in December (manual task E), hand them over, and I add the tables. Then 🏛️ UPDATE TAX RATES in early January and ♻️ RESCORE ALL.

Medium

4 · The s.r.o. tax model misses three rules MEDIUM

`config.TAX_YEAR_RULES[2026]`, `engine/financial.py` (`calc_tax_sro`, `analyse`, `project_irr`) · účtovník + code · S

Problem

(a) Dividend tax. The table has 10 %. Slovak guidance says 10 % applied to profits of 2024 only, and profits of tax periods from 1 January 2025 are taxed at 7 % again.
(b) Minimum tax. A company pays at least €340 a year when its income is up to €50,000, also when it makes a loss. The model charges the computed tax only: €28 a year for a Petržalka flat, €0 for a cheap one.
(c) Depreciation. A company depreciates a flat over 40 years (group 6), up to the rent it earns from it. On a €150,000 flat that is up to €3,750 a year off the taxable profit. The model deducts none. The code's own comments say so for Czechia.

Impact

The s.r.o. figures are off by a few hundred euros a year, in either direction: too favourable for a heavily mortgaged flat (the €340 minimum), too harsh for one bought mostly with cash (no depreciation, 10 % dividend). The recommendation is Personal almost everywhere, so few verdicts change; the numbers under the s.r.o. toggle do.

Fix

Ask your účtovník to confirm all three (manual task C), then: dividend rate 0.07, tax = the larger of the computed tax and the minimum, depreciation as price ÷ 40 capped at the year's rent. Raise the model version so every flat is re-scored.

5 · The dashboard shows the s.r.o.'s figures by default MEDIUM

`dashboard/sidebar.py:259` (`value=True`), `view.py`, `cards.py`, `triage.py:89`, `compare.py:61`, `memo.py:295` · code · S · ✔ seen in the browser

Problem

“Show s.r.o. figures” is on when the page opens. So the surplus on every card and in the triage table, the ⛔ cash-flow flag, the “cash-flow positive only” filter and the cost breakdown are the company's — next to a badge that says PERSONAL is recommended. The +2 pp stress figure is the s.r.o.'s whatever the toggle says. The Telegram alert uses the better of the two structures, so a flat can be alerted and flagged negative on the dashboard.

Fix

Default to the **recommended** structure per listing (a third toggle position, “Recommended”), store the stress result for both structures, and use the same rule for the flag, the filter and the alert.

6 · The duplicate matcher merges different flats MEDIUM

`engine/duplicates.py` (`is_same_flat`, `group_duplicates`) · code · S–M · ✔ reproduced

Problem

Two listings are one flat when the district, size (±1 m²), price (±3 %), rooms and floor agree, and an unknown floor agrees with anything. Groups are then chained. Panel-built districts are full of flats with the same floor area at near-identical prices.

Evidence

Petržalka: 68.0 m², €189,000, floor unknown · 68.6 m², €192,500, floor 7 · 67.5 m², €187,000, floor 2. All three came out as one flat, although two of them state different floors.

Impact

With “hide duplicates” on, two real listings disappear behind the third. A failed LV on one marks the others “a copy failed LV”.

Fix

Never chain through a member that contradicts another (compare every pair in a group). Add evidence that two ads are one flat: map pins within about 100 m when both have one, the same street, the same first photo. With none of these, require an exact size match.

7 · On a Mac without Docker nothing runs by itself MEDIUM

`Start Sovereign.command` · code · S

Problem The Mac launcher starts the dashboard and nothing else. The 06:00 pipeline, the backup that is its first step and the Telegram alerts happen only when you click ▶ RUN EVERYTHING, and only while the Terminal window is open.

Fix Start the scheduler from the launcher alongside the dashboard, or install it as a macOS launch agent that starts at login. Until then use manual task H.

8 · An LV PASS never expires, and its date isn't shown MEDIUM

`database/title_deeds.py` (`get_pending_lv`), `kataster_scraper.py` (`CACHE_TTL_DAYS = 30`), `dashboard/cards.py` · code · S

Problem The daily check re-reads PENDING deeds and some UNVERIFIED ones. A PASS stays a PASS for as long as the listing lives, and the check that gave it may have read a copy of the deed up to 30 days old. The date of the check is stored (`lv_checked_at`) but no card shows it. An exekúcia entered last week is invisible until you press re-verify.

Fix Show “LV PASS · checked 3 Nov” on the card and in the memo. Re-check PASS deeds of deals in OFFER or later every week, and any PASS older than 30 days before it is shown as verified.

9 · Some banks aren't recognised MEDIUM

`modules/lv_screen.py` (`_BANK_RE`), `config.LV_BANK_NAMES` · code · S · ✔ reproduced

Problem A lien is ordinary when its creditor is a bank. The rule needs “banka” or “bank” as a separate word, or a name on the list. One-word names fail: **Privatbanka**, and names that old liens still carry: Istrobanka, Volksbank, Sberbank, Citibank. So does the abbreviation **SLSP**.

Impact The safe direction, but a lost deal: the flat is REJECTED for an ordinary mortgage. Claude can clear it when a key is set.

Fix Accept “…banka” and “…bank” as a word ending, and add SLSP, Privatbanka and the former names to the list, with tests.

10 · Villages and city parts outside the lookup have no benchmark MEDIUM

`engine/regional_prices.py` (`kraj_for_district`), `engine/financial.py` (`match_rent_key`) · code · M · ✔ reproduced

Problem A place the lookup doesn't know gets no regional median, so it is WHITE at any price, and the national default rent of €6.50/m². Dunajská Lužná (a commuter village by Bratislava), Košická Nová Ves (a Košice borough), Trenčianske Teplice and Veľký Meder all came out that way.

Fix Resolve every listing to its okres: from the geocoder's answer (it carries the county), or from the Statistical Office's free list of the roughly 2,900 municipalities with their okres. Then the benchmark and the kraj rent apply everywhere.

11 · Backups on the same disk; restore has no safety copy MEDIUM

`modules/backup.py` · you · S, code · S

Problem Open since the last audit: a disk failure or theft takes the database and all 14 backups together. New: `restore_database` writes a backup over the live database without saving the live one first, so restoring the wrong file can't be undone. Backups are never test-opened.

Fix You: sync `backups/` off the machine (manual task F). Code: save the live database before a restore, and run SQLite's integrity check on each new backup.

12 · Seven Slovak towns still use the property-tax fallback MEDIUM

TAX RATES tab · you · S

Problem Martin, Považská Bystrica, Nové Zámky, Levice, Lučenec, Topoľčany and Rimavská Sobota. Their flats are charged €1.35/m². I could not find their 2026 ordinances online from here.

Fix Manual task G. Section 8 also lists a way to read scanned ordinances with the Claude key you already have.

Low and Info

#	Finding	Detail and fix
13	"Novostavba" means developer project	✔ "Novostavba 2-izb. byt s balkónom" was flagged, and "Hide dev projects" is on by default. The flag never comes off once set. Fix: require a second sign (the project URL, "od €", a phase or stage number), and clear the flag when the signs go.
14	CSV formula injection	✔ `=HYPERLINK(…)`, `+1+1` and `@SUM(1)` reached the file unchanged. Fix: put an apostrophe before any text cell that starts with `=`, `+`, `-` or `@`.
15	The map's opening view	✔ One test listing with coordinates outside the country moved the centre to western Ukraine. Fix: centre on the median, drop pins outside the market's box when drawing, and validate coordinates where they are saved, not only in the scrapers.
16	Telegram message length	Deals are capped at 10; cuts and problems are not. About 25 tracked cuts in one run would pass 4,096 characters. Fix: cap each section and say "…and N more". Also read the sent-cuts list once, not once per cut.
17	Size or rooms corrected	`upsert_listing` drops a score when the price or district changes. A portal that corrects 55 m² to 65 m² leaves the old score. Fix: drop it for size and rooms too.
18	Legacy Google Places endpoint	`maps/api/place/nearbysearch/json` is the legacy API. Fix: move to Places API (New), or drop the Google path and keep OpenStreetMap, which already works without a key. Check in the Google Cloud console which one your project can enable.
19	District `None`	✔ `is_industrial_zone` calls `.lower()` on it. Fix: `(district or "")`.
20	Login	Fine on 127.0.0.1. Before opening it to a network use a private tunnel (section 8, row 15) rather than a bare port.
21	Tall header	Fix: collapse the banner to one line with "details", put tiles and the health strip on one row, and remember the last tab.
22	Košice - Staré Mesto	✔ The Bratislava city-part rule is skipped (correctly), and the city anchor then picks Košice II. Fix: map Košice's boroughs to Košice I–IV as Bratislava's are.
23	Slovak rate and term not in `.env`	Fix: `MORTGAGE_RATE_PA` and `LOAN_TERM_YEARS` as settings, like `CZ_MORTGAGE_RATE_PA`. Section 8 lists a free source that could keep the rate current by itself.
24	Slovak cooperative flats	Fix: detect "družstevný byt", "členský podiel", "prevod členských práv" in the ad as the Czech check does, mark the listing, and never call its deed clean.

25	Ad text to Claude	The answer is limited to a fixed set of fields. No change needed; noted for completeness.
26	Pasted credentials	Manual task A.
27	The market's yields	See finding 2's table. Worth knowing when you read the tiles: GREEN means cheap against the median, not self-funding.
28–29	Broad excepts, pandas warnings	Nothing to do.

5. What needs manual checking — only you can do these

No code change replaces these. They need a decision, a professional, an account, or your own computer.

Task	Why	When	Time
A. Revoke the two pasted credentials	Finding 26	Today	10 min
B. Fill in `.env`	Alerts, Claude's second reading, your own tax position	Today	20 min
C. Účtovník: confirm the tax model	Finding 4, and three items from the last audit	This month	1 h + adviser
D. Lawyer or notary: confirm the LV rules and the contract draft	Finding 1, and the § 15 lien rule applied in PR #76	Before your next offer	1 h + adviser
E. Collect the 2027 figures	Finding 3	1–20 December, then early January	2–3 h
F. Off-site backup, and one test restore	Finding 11	This week	30 min
G. Upload seven towns' tax ordinances	Finding 12	This month, again in January	1 h
H. Run the scheduler on your Mac	Finding 7	Until it is fixed in code	5 min
I. Put your own financing into the model	Findings 2, 23, 27	Before you trust a surplus	1–2 h
J. Per deal, every time	Things no model sees	Before an offer and before signing	1–2 h per deal
K. Weekly health check	Dependabot, portals	Weekly	10 min
L. Legal advice before selling access	Unchanged from the last audit	Before any launch	Adviser

A · Revoke the two credentials pasted into this chat

Resources: Telegram `@BotFather` · console.anthropic.com · claude.ai → Settings

1. Telegram: open `@BotFather`, send `/revoke`, pick your bot. It gives a new token; the old one stops working at once.
2. Put the new token in `sovereign_final/.env` as `TELEGRAM_BOT_TOKEN=…`. Your chat id stays the same.
3. Anthropic: the text you pasted starts with `sk-ant-usr-`. That is a sign-in credential for your account, not an API key. Sign out of all sessions in your Claude account settings, or revoke it wherever it was issued.
4. Create a real API key at console.anthropic.com → API Keys. It starts with `sk-ant-api03-`. Set a monthly spend limit under Billing.
5. Never paste either into a chat, an email or a file inside the repository. `.env` is the one place; git ignores it.

B · Fill in `.env`

Resources: `sovereign_final/.env.example` (each setting is explained there) · registrace.cuzk.gov.cz

1. Copy `.env.example` to `.env` if you haven't.
2. Set `TELEGRAM_BOT_TOKEN` (new, from task A) and `TELEGRAM_CHAT_ID` (the number you saw in the browser when you set the bot up).
3. Set `ANTHROPIC_API_KEY` (from task A). This turns on Claude's second reading of every title deed.
4. Set `OTHER_ANNUAL_INCOME` to your yearly tax base from everything else, in euros. With €50,000 the rental is taxed at 25 %, not 19 %.
5. Set `PROPERTIES_OWNED` (2 or more switches to the 70 % loan cap) and `NOMINATIM_EMAIL`.
6. For Czech deals: a free ČÚZK key in `CZ_CUZK_API_KEY`.
7. Until finding 2 is fixed: `CASHFLOW_MIN_SURPLUS=-200` lets deals within €200 a month of break-even through to Telegram.
8. Restart the dashboard. Click 📨 TEST TELEGRAM under DIAGNOSTICS, then ♻️ RESCORE ALL.

C · Účtovník: confirm the tax model

Resources: skdp.sk (Slovak Chamber of Tax Advisers) · podpora.financnasprava.sk · slov-lex.sk (Act 595/2003)

1. Send these questions as they are:

```
PERSONAL (rental under § 6 ods. 3, no živnosť)
1. 2026 bands: 19 % to €43,983.32, 25 % to €60,349.21, 30 % to €75,010.32,
   35 % above — correct? Subsistence minimum €284.13?
2. Is interest on the mortgage that bought the rented flat deductible?
   Which other costs are (property tax, insurance, fond opráv, repairs)?
3. How exactly is the €500 exemption applied?
4. Health contributions on this income in 2026: none?
5. Sale after 5 years: exempt?

S.R.O. (one flat, rent about €8,000 a year)
6. Dividend withholding tax on profit of 2026: 7 % (not 10 %)?
7. Minimum tax (daňová licencia): €340 a year at income up to €50,000,
   also in a loss year?
8. Depreciation of the flat: group 6, 40 years, straight-line, capped at
   the rental income of the year? What about the land share?
9. Corporate rate 10 % up to €100,000 income, 21 % above?
10. Does the financial transaction tax apply to the company's payments
    (loan instalments, fond opráv)? How much a year, roughly?
11. Can I lend the company my own cash and take it back as loan
    repayments rather than dividends?
```

2. Send me the answers, or write them next to `TAX_YEAR_RULES[2026]` in `config.py` with the date and the adviser's name.
3. If interest is deductible: `PERSONAL_INTEREST_DEDUCTIBLE=1` in `.env`, then ♻️ RESCORE ALL.
4. Buying in Czechia: the same with a daňový poradce (§ 9 flat rate against actual costs, SVJ payments, the 10-year test).

D · Lawyer or notary: the LV rules and the contract draft

Resources: notar.sk · sak.sk · kataster.skgeodesy.sk · Act 182/1993 on slov-lex.sk

1. Bring three to five real LVs of flats you looked at.
2. Ask:

```
1. § 15 of Act 182/1993 statutory lien of the other owners: the dashboard
   now shows it as a warning, not a rejection, and tells me to get a
   potvrdenie o bezdlžnosti from the správca. Is that right?
2. Transfer bans (zákaz scudzenia / prevodu, obmedzenie prevodu): a bank's
   is a warning, anyone else's rejects. Right?
3. Which of these should always stop a purchase, and which only warn?
   plomba / pending vklad · právo doživotného užívania · duplicitné
   vlastníctvo · spochybnená hodnovernosť · obmedzenie nakladania ·
   nútená správa · vyvlastňovacie konanie · reštrukturalizácia ·
   právo spätnej kúpy · nájomné právo · zádržné právo
4. Is there anything else on a Slovak LV you would never accept?
5. Please review the kúpna zmluva draft (ONE-CLICK CLOSE).
```

3. Give me the answers; question 3 is exactly the list finding 1 needs.

E · Collect the 2027 figures (every December)

Resources: slov-lex.sk · financnasprava.sk · nbs.sk · cnb.cz · financnisprava.cz · zakonyprolidi.cz

1. 1–10 December: on slov-lex.sk open Act 595/2003 in the version effective 1.1.2027. Note the bands, the subsistence minimum, corporate and dividend rates, the minimum tax.
2. nbs.sk: the loan-to-value caps (80 %, 70 % for a third property) and whether the 1 October 2026 start date held. cnb.cz: the Czech limits.
3. A current average mortgage rate for both countries (section 8, row 1).
4. Czechia: the 2027 inflation coefficient for property tax, and the 2027 average wage that sets the 23 % threshold.
5. Hand me the numbers with their sources; I add the 2027 tables.
6. 2–10 January: 🏛️ UPDATE TAX RATES, review doubtful readings, ♻️ RESCORE ALL. The “stale tax year” notice should be gone.

F · Off-site backup, and one test restore

Resources: iCloud Drive, Google Drive or an external disk · README “Backups”

1. Mac: run `crontab -e` and add one line (adjust the first path):

```
30 7 * * * rsync -a ~/path/to/sovereign_final/backups/ ~/Library/Mobile\
Documents/com~apple~CloudDocs/Sovereign-backups/
```

2. The backups hold title-deed texts with owners' names. Use a private folder, not a shared one.
3. Test once. First copy `data/sovereign.db` somewhere safe by hand (finding 11), stop the scheduler, then:

```
cd ~/path/to/sovereign_final
.venv/bin/python -m modules.backup restore backups/sovereign-YYYY-MM-DD.db
```

4. Open the dashboard and check your deal stages and notes.

G · Upload the seven towns' tax ordinances

Resources: each town's site (“VZN daň z nehnuteľností 2026 <town>”) · cuet.slovensko.sk

1. Click 🏛️ UPDATE TAX RATES from your home connection first; town sites often block data-centre addresses only.
2. For each town still on the fallback, download its VZN as a PDF.
3. 📊 ANALYSIS → TAX RATES → the town → upload. Accept the reading when the rate per m² for “byty” matches the PDF.

H · Run the scheduler on your Mac (until finding 7 is fixed)

Resources: Terminal

1. Start the dashboard once with `Start Sovereign.command` so its `.venv` folder exists.
2. Open a second Terminal window and run:

```
cd ~/path/to/sovereign_final
caffeinate -i .venv/bin/python scheduler.py
```

3. Leave the window open. `caffeinate` keeps the Mac awake while it runs, as long as the lid is open. If the Mac was asleep at 06:00, the run starts when it wakes. A run missed since the last 06:00 starts at once.
4. Lines are also written to `logs/scheduler.log`.

I · Put your own financing into the model

Resources: two banks' offers · your účtovník · the WHAT-IF tab

1. Ask two banks for a real offer on a flat like the ones you look at: rate, fixation, term, maximum loan share, fees.
2. Until finding 23 is fixed, the Slovak rate and term live in `config.py` (`MORTGAGE_RATE_PA`, `LOAN_TERM_YEARS`); tell me the numbers and I set them. A 30-year term alone moves break-even by about five points (finding 2's table).
3. For an s.r.o.: ask the bank's business desk for the rate, loan share and whether you must guarantee personally. Put them in `.env` as `SRO_RATE_PREMIUM_PP`, `SRO_LTV_RATIO`, `SRO_ANNUAL_RUNNING_COST`.
4. Check the model's other assumptions against what you know:

Assumption	Value	Check against
HOA incl. fond opráv	€35 / €60 / €90 / €130 by size	the building's own monthly statement
Vacancy · reserve	5 % · 5 % of rent	your experience; an agent
Buying costs	1 % of the price	notary, cadastre and bank fees quoted to you
Price growth · rent growth	3 % · 2.5 % a year	NBS and Deloitte reports
Selling costs	3.5 %	an agent's commission
s.r.o. set-up · running	€2,500 · €1,200 a year	your účtovník's quote
Rent table	April–May 2026	three live rental ads near the flat
Czech figures marked UNVERIFIED	README lists them	a Czech adviser

J · Per deal, every time

Resources: the seller's agent · the building's správca · kataster.skgeodesy.sk · cre.sk · justice.gov.sk · Czechia: nahlizenidokn.cuzk.cz, isir.justice.cz

1. Get the flat's LV number and cadastral area from the agent. Enter it in WORKFLOW → LV TO-DO and verify.
2. **Read part C yourself**, whatever the badge says (finding 1). Anything you don't understand goes to the lawyer.
3. Ask the správca in writing for: a potvrdenie o bezdlžnosti for the flat, the fond opráv balance, loans the building has taken, and works planned for the next three years.
4. Check the seller: insolvency register (free), and the register of executions (a small fee per lookup). In Czechia: ISIR and the Central Register of Executions.
5. Ask for the energy certificate and the last year of utility bills.
6. Find three live rental ads within a few streets and compare with the card's rent.
7. View it. Note the building's state: roof, façade, lift, risers.
8. Within 48 hours of signing: press re-verify LV again (finding 8), and have the lawyer read the fresh deed.

K · Weekly health check

1. GitHub: merge Dependabot pull requests whose checks are green. Then `git pull origin main` and restart.
2. Dashboard header: no red banner; no red chip in the health strip. A portal red for two days: click 🔗 TEST SITES and tell me what it says.
3. Telegram: you should get a “failed run” message when a run fails. If you get nothing for a week, check that the scheduler window is still open (task H).

L · Before selling access or assigning deals for a fee

1. Written legal answers on: reuse of scraped portal data, the unofficial cadastre scraper, GDPR (owners' names), brokerage licensing, and whether a ranked paid feed is investment advice. Unchanged from the last audit.

6. What is missing

Gaps between what the tool does and what a buyer needs from it. Each one is a hole today, not a nice-to-have.

#	Missing	Why it matters	Closes with
M1	A safety net for title-deed entries the Slovak screen doesn't know	The one check that must not fail open does	Finding 1
M2	Alerts that match how you invest	The bot stays silent on deals	Finding 2
M3	Running by itself on your Mac	Stale data, no backup, no alerts	Finding 7
M4	A check that the daily run happened	If the scheduler stops, nothing tells you; silence looks the same as “no deals”	Section 8, row 16
M5	Your own financing as settings	Every surplus uses one average rate and a 25-year term	Finding 23

M6	Sold prices	Every benchmark is an asking price; the gap to what flats sell for is unknown	Section 8, rows 7–8 (partly)
M7	Seller checks	An insolvent seller or one under execution is found only if it is already on the deed	Section 8, row 6
M8	A due-diligence checklist per deal, with its documents	Task J lives in this PDF, not in the tool; LV PDFs, certificates and contract versions have no home	Suggestion S3
M9	The building's own facts	Age, construction type, whether the building is renovated, heating, real HOA fee: the biggest cost surprises, and none is modelled	Suggestions S4, S12
M10	Rent by flat type	One rent per m² per district, adjusted by a fixed room factor; live comps aren't split by rooms or furnishing	Suggestion S9
M11	Life after the purchase	The tool stops at the contract draft: no real rent, costs, tax or loan balance of flats you own	Suggestion S10
M12	Safe access from your phone away from home	The dashboard is bound to the one computer; opening the port is not safe	Section 8, row 15
M13	Slovak cooperative flats, and auctions as a source	The first can look like bargains; the second are left out on purpose and are where below-market flats often are	Finding 24, section 8, row 14
M14	A yearly reminder inside the tool	Finding 3 comes back every December	Suggestion S14

7. What could be added

Ranked by what they would change for you. **S** = under half a day, **M** = a few days, **L** = a week or more. Items from the last audit that are still open keep their old number in brackets.

#	Suggestion	Why	Effort
S1	**Alert rules you set**: by class, grade, district, size, maximum price, minimum surplus; several saved searches, each with its own alert (N10)	Fixes finding 2 properly and lets you watch a target area	M
S2	**“Read part C” on every card**: the deed's encumbrance section as text, with the screen's hits highlighted and the unknown lines marked (N16)	Makes task J step 2 one click; shows you what the screen saw	S
S3	**Due-diligence checklist per deal** with file upload: LV, potvrdenie o bezdlžnosti, energy certificate, bank offer, contract versions (N3)	The steps that protect you most are the ones outside the tool	M
S4	**Správca letter**: a pre-filled request for arrears, fund balance, building loans and planned works (N4)	The building's debts are invisible on the deed	S
S5	**“Your situation” panel** in the sidebar: other income, properties owned, your rate, term and loan share, saved to `.env` (N1)	Findings 5 and 23; no file editing	M
S6	**Mortgage-offer comparer and fixation-end shock**: enter two or three offers; model the rate resetting after 3, 5 or 10 years (N5, N6)	That is how Slovak and Czech loans reprice; +2 pp from day one is cruder	M
S7	**“Why this grade?”** on each card: points from yield, discount, location, energy, condition, risk (N9)	A is a black box today; a flat with no location data can score 83	S
S8	**Offer sheet**: max offer, the comparables behind it, price history, days on market, as one PDF page (N2)	Turns “max offer” into an argument at the viewing	M

S9	**Rent comps by rooms and furnishing**, and a “time to let” estimate from how long rental ads stay up	Missing item M10; the data is already scraped	M
S10	**Portfolio view** for flats you own: real rent, costs, tax, loan balance, equity (N11)	Also calibrates the model's assumptions with your real numbers	L
S11	**Backtest**: of last month's GREENs, how many disappeared, how fast, after what cuts (N12)	Tells you whether GREEN means cheap; a first step towards sold prices	M
S12	**Renovation and heating-cost estimates** by condition, size and energy class; a flip scenario (N8, N17)	Many GREEN flats are GREEN because they need work	M
S13	**Short-term let scenario** for the Tatras, Bratislava I and central Prague (N7)	Poprad's rent already carries a hidden Airbnb premium	M
S14	**December checklist page** and a weekly digest on Telegram or email (N13, N15)	Finding 3 recurs; a digest shows the tool is alive	S
S15	**Email and push alerts** next to Telegram	A second channel when Telegram is muted	S
S16	**Travel time on the map** (walk, transit, drive) to a point	Straight-line distance misleads across a river or a hill	S–M
S17	**Read scanned documents with Claude**: tax ordinances that are scans, a photographed LV from an agent	Clears finding 12 and the 231 unreadable Czech ordinances with the key you have	M
S18	**“Sold or withdrawn” notice** for tracked deals, and days-on-market per district	Tells you how much time you have to decide	S

8. APIs and data sources that would improve a feature

✔ = confirmed by a web search during this review. **check** = from general knowledge or a single source; confirm on the site before building on it. Limits and prices change; the sites are the authority.

#	Feature it improves	API or source · where to get it	Cost · key	
1	**Mortgage rate kept current** (today 3.8 % and 5.51 % are typed in)	**ECB Data Portal**: series `MIR.M.SK.B.A2C.A.R.A.2250.EUR.N`, the monthly rate on new housing loans in Slovakia · data.ecb.europa.eu **ČNB ARAD REST API** for Czech rates · cnb.cz/arad (make an account, generate a key in your profile)	Free, no key Free, key	✔ series and ARAD · **check** the ECB REST address
2	**Travel time and "reachable in 20 minutes" areas on the map**	**openrouteservice** · openrouteservice.org → sign up → request a key **Mapy.com REST API** · developer.mapy.com (Seznam account) **Google Routes API** · console.cloud.google.com	Free: about 2,000 routes and 500 area maps a day Free: 250,000 credits a month; a route costs 4 10,000 free calls a month per product, then paid	✔
3	**Better address-to-pin** for Slovak and Czech addresses (today OpenStreetMap's public server, or Google)	**Mapy.com geocoding** (strong in both countries) · as above **Google Geocoding** · as above Czechia: **RÚIAN** address points from ČÚZK · cuzk.gov.cz	Free tier 10,000 free a month Free	✔ Mapy, Google · **check** RÚIAN
4	**Transit and amenities**	Stay on OpenStreetMap (works now, no key). If you want Google: **Places API (New)**, not the legacy one (finding 18). City timetables: Bratislava and Prague publish theirs as open data	Free Free	**check**
5	**Title deeds read by machine**	Slovakia: **none**. The cadastre office offers map services, no ownership API; the app's scraper is the only route. An official LV for a contract comes from the cadastre office, its e-services or a notary. Czechia: **ČÚZK API KN** (in use, free key) and **WSDP remote access**, which returns the full LV · cuzk.gov.cz → Dálkový přístup	— API KN free; WSDP paid per output	✔
6	**Seller checks** once the deed names the owner	Czechia insolvency: **ISIR web service** · isir.justice.cz Czechia companies: **ARES REST API** · ares.gov.cz Slovakia insolvency: the Ministry of Justice's new register of insolvency proceedings · justice.gov.sk (search page; I found no public API) Slovakia executions: **Centrálny register exekúcií** · cre.sk (no API) Slovakia companies: **Register účtovných závierok** API · registeruz.sk	Free Free, no key Free About €1.60–2.50 per lookup Free	✔ ISIR, ARES, CRE fees · **check** the rest

7	**Price benchmarks kept current**	**NBS** quarterly asking prices by kraj · nbs.sk → Statistics (file download; I found no API) **Statistical Office DATAcube** · datacube.statistics.sk **Eurostat API** (house price index, both countries) · ec.europa.eu/eurostat Czechia: **ČSÚ open data** (in use)	Free	**check**
8	**Rents**	Czechia: Ministry of Finance rent map (in use). Slovakia: no public rent API; your own scraped comps are the best source. Paid Czech valuation services exist; ask for a trial before paying.	Free	✔ MF map
9	**Short-term let scenario**	**Inside Airbnb**: free quarterly files; Prague is covered, check for Bratislava · insideairbnb.com **AirDNA** · airdna.co	Free Paid, price on request	✔ Prague · **check** Bratislava
10	**Scanned ordinances and photographed LVs**	**Claude API**, which you already use: it accepts PDF pages and images · console.anthropic.com	Paid per use; your existing key	**check** current prices
11	**Alerts beyond Telegram**	Email through your own mailbox (an app password), or **ntfy** for phone push · ntfy.sh	Free	**check**
12	**Exchange rate fallback**	ČNB daily file (in use). Fallback: **ECB reference rates** · data.ecb.europa.eu	Free	**check**
13	**Risk layers**	Flood: SVP map (in use). Noise: the EU strategic noise maps from the **European Environment Agency** · eea.europa.eu. Air quality: **OpenAQ** · openaq.org	Free; OpenAQ needs a key	**check**
14	**Auctions as a deal source**	Slovakia: the notaries' central register of auctions · notar.sk Czechia: the executors' auction portal · portaldrazeb.cz Both are web pages, no API; they would need their own reader and a legal look at their terms.	Free to read	**check**
15	**Safe access from your phone**	**Tailscale** (a private network between your devices) · tailscale.com **Cloudflare Tunnel** with its own login · cloudflare.com	Free for personal use	**check**
16	**“Did the daily run happen?”**	**Healthchecks**: the pipeline pings a web address after each run; you get an email when a ping is late · healthchecks.io	Free tier	**check**
17	**Village → okres** (finding 10)	The Statistical Office's list of municipalities with their okres; or the county field the geocoder already returns	Free	**check**
18	**Portals that block the scrapers**	ScraperAPI (already supported) · scraperapi.com. No Slovak or Czech sale portal offers a public listings API; Sreality's stays off because its robots.txt forbids it.	Free tier, then paid	—

Where I would start: row 1 (free, removes a typed-in number), row 16 (ten minutes, closes missing item M4), row 2 with openrouteservice (free, finishes the map), and row 10 (uses a key you already pay for).

9. What happened to the previous audit (6 Nov 2026)

23 of the 26 actionable findings are fixed in pull requests #74, #75 and #76, all merged. Three stay open, and all three need you; three were notes. **All 8 open dashboard ideas are done.** Of the 17 new suggestions, 1 is done and 1 partly.

#	Finding (6 Nov)	Status	Notes
1	LV screen misses transfer bans	FIXED	PR #76. The wider gap is this audit's finding 1.
2	2026 Slovak tax bands	FIXED	Four bands, on top of `OTHER_ANNUAL_INCOME`. Účtovník to confirm (task C).
3	Tax tables expire 1 Jan 2027	OPEN	Finding 3
4	§ 15 owners' lien rejects good flats	FIXED	Now a warning. Lawyer to confirm (task D).
5	s.r.o. compared at different leverage	FIXED	Extra cash is charged its cost
6	Mortgage interest not deducted	FIXED	As a switch, off until your účtovník confirms
7	Headline returns always the s.r.o.'s	FIXED	The toggle's default is this audit's finding 5
8	Hold slider didn't move the verdict	FIXED	
9	Czech towns unscored	FIXED	Ministry of Finance rent map, about 4,800 municipalities; runs 15–25 % under Deloitte
10	Keyword-only LV checks without a key	FIXED	The dashboard says so. The key is still yours to set (task B).
11	No off-site backup	OPEN	Finding 11, task F
12	LV texts kept for ever	FIXED	Deleted after 180 days
13	Seven towns on the tax fallback	OPEN	Finding 12, task G
14–26	Thirteen low findings	FIXED	PR #74: process-tree kill, 0 % loans, IRR "—", blank cells, Deploy hidden, log rotation, Docker non-root, contract wording, bank s.r.o. exclusion, schema checks, batched queries, rent 0, login lockout
27–29	Three notes	NOTED	

Dashboard idea (6 Nov)	Status
Telegram alerts	DONE PR #75 — see finding 2 for the rule
CSV export · column chooser · side-by-side comparison · portal health strip · richer map · "last seen" · Deploy hidden · compact phone cards	DONE PRs #74 and #76. The map filters by straight-line distance, not travel time (section 8, row 2). The export is CSV, not XLSX.
N14 keyword-only notice	DONE
N1 "your tax situation"	PARTLY the setting exists; no panel (S5)
N2–N13, N15–N17	OPEN carried into section 7

10. How this review was done, and sources

- **Tests and static checks:** the full suite on Python 3.13; ruff with CI's rules and extra security, exception and date rules; vulture; pip-audit; a secrets search over the files and all 263 commits.
- **Reading the code:** the financial engine, config and markets, the Slovak and Czech LV screens and the decision logic, alerts, the scheduler, the database layer for listings and title deeds, duplicates, backup, login, the launchers and Docker files.
- **Probes:** 28 engine edge cases, 33 crafted title-deed entries, look-alike flats for the duplicate matcher, the CSV export, and the break-even discount in six towns.
- **Running the app:** a scratch copy with 221 listings in both markets, 11 of them hostile (script tags in every text field, missing fields, a €9.9 billion price, coordinates outside the country). Chromium clicked all 12 tabs at 1440 px and 390 px, opened cards, switched markets, used the pickers and the map.
- **Web checks:** the Slovak s.r.o. rules and each API in section 8.

Not covered: live scraping of the real portals and the cadastre (it would hit third-party sites, and the review machine sits in a data centre they block), your real database, and a run on Windows or macOS (CI covers Windows).

Sources

- podnikajte.sk — Zrážková daň z dividend v roku 2024 a 2025
- podnikajte.sk — Sadzby dane z príjmov právnických osôb a daňová licencia v roku 2026
- Finančná správa — Všeobecne o minimálnej dani
- podnikajte.sk — Odpisové skupiny v roku 2026 · 10 faktov o odpisovaní majetku
- ECB Data Portal — MIR.M.SK.B.A2C.A.R.A.2250.EUR.N
- ČNB — ARAD REST API
- openrouteservice — plans · restrictions
- Mapy.com — REST API pricing
- Google Maps Platform — pricing overview
- ČÚZK — Remote access to cadastre data · ÚGKK SR — web map services
- Ministerstvo financí ČR — ARES
- Ministerstvo spravodlivosti SR — Register predinsolvenčných, likvidačných a insolvenčných konaní
- podnikajte.sk — Centrálny register exekúcií: ako funguje a koľko stojí
- Inside Airbnb
- Previous audit: `sovereign_final/docs/AUDIT_REVIEW_2026-11-06.pdf`

These are secondary sources read through a search. Tax and legal points need your účtovník's and lawyer's confirmation (tasks C and D).